

A generic planning approach for sustainable supply chain management - How to integrate concepts and methods to address the issues of sustainability?



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ABSTRACT

Forced by competitive, legislative and customers' pressures, corporate executives are supposed to consider sustainability aspects of value creation which induces a new set of challenges within decision-making. Unfortunately, guidelines supporting comprehensive analysis, especially concerning the evaluation of environmental and social performance of decision alternatives, are missing. This hinders the advancements in corporate sustainability. Questions arise on how to measure and balance respective indicators with traditional economic objectives. To fill this void, this research sets out to develop an intuitive and holistic planning framework for advancing sustainability along supply chains. In a first step, we identify sustainability issues and indicators as well as decision-support methods by scrutinizing literature on sustainable supply chain management. Based on this, we develop a planning framework that supports executives in evaluating and deciding upon corporate sustainability initiatives. The framework comprises three consecutive phases of planning: 'Problem definition', 'Assessment of alternatives', and 'Decision analysis'. For each phase we propose appropriate tools distinguishing concepts of sustainable supply chain management (e.g. Sustainable Logistics and Sustainable Manufacturing), methods for measuring the ecological and social efficiency of decision alternatives, and quantitative decision-support techniques able to balance trade-offs between the three dimensions of the triple bottom line. As a result, this article presents a framework for guiding corporate decision-processes. From the academic angle, the article contributes to the research field by providing a structured integration of concepts and methods utilizable in the pursuit of ecological and social improvements along the value chain.

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1. Introduction

Societal and competitive pressures as well as legislation elevate the attention companies pay to environmental and social effects of their supply chain activities (Meixell and Luoma, 2015). According to business practice, known as sustainable supply chain management (SSCM), mitigates risks (Giannakis and Papadopoulos, 2016), but may also act as a differentiator (Jain, 2012). The risks are exemplified by media reports on unsocial labor practice, as observed at Foxconn, the Rana Plaza in Bangladesh, or incidents of child labor at Nike's supply chain (Meixell and Luoma, 2015; Sancha et al., 2016; Wolf, 2014). As a differentiation strategy, companies like Subaru and Toyota aim for zero-waste facilities (Hassini et al., 2012) and

follow the idea of zero-emission warehouses (Dekker et al., 2012). Favoring a pro-active strategy, sustainable business practice is even considered to be a basic requirement for market participation (Wolf, 2014) and a critical success factor (Kushwaha and Sharma, 2016).

Undisputably, the integration of sustainability "is one of the most important subjects in the modern study of operations management" (Jabbour and de Sousa Jabbour, 2016, p. 1827), with research on SSCM being on the rise (Ashby et al., 2012). Nevertheless, we have to acknowledge a "persistent gap between the diffusion of sustainability discourse and its practical application" (Ashby et al., 2012, p. 509). Indeed, practitioners hesitate to embark into SSCM as it "remains an extremely confused and complex issue" (Ageron et al., 2012, p. 172) with sustainability practices in SCM being hard to understand (Pimenta and Ball, 2015). The obstacles may be largely attributed to a lack of guidelines on how to measure,

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pursue and actually implement SSCM within business practice (Fabbe-Costes et al., 2014; van Wassenhove and Besiou, 2013). This is in line with Schaltegger and Burrit (2014) who concede a need for a structured framework that decision-makers can use to advance sustainability within their supply chain. This research sets out to fill this void by developing a planning framework for SSCM. Taking the above mentioned observations into account, the respective development process is guided by three research questions:

- Q1.** What are appropriate metrics to measure sustainability within supply chain processes?
- Q2.** Which concepts and methods for facilitating the design and implementation of sustainable supply chains are presented in the academic discourse?
- Q3.** How can these concepts and methods be integrated within a comprehensive planning process considering the measures identified in Q1?

Overall, this study develops an intuitive and holistic framework that may benefit practitioners to detect potential fields for corporate sustainability initiatives and to provide them with a proper methodological toolset for evaluation. In this way, we hope to take another step to bring "notions of sustainability [...] into the boardrooms of companies" (Schaltegger and Burrit, 2005, p. 187) by easing the integration of sustainability aspects into corporate decision-making processes. From an academic angle, the framework opens new pathways in research on SSCM by presenting various concepts and methods in an integrated manner. To do so, this research is scrutinizing literature positioned in the field of SSCM, identifies existing concepts that facilitate the actual implementation of this paradigm and provides methods for evaluation of SSCM initiatives along the value chain.

2. Definitions and research gap

In this section we define the scope of our research and our understanding of basic terms. Moreover, we identify the research gap as a starting point for our research and delimit this study from extant literature.

2.1. Sustainable supply chain management

SSCM is a key enabler for advancing sustainable development in the industrial sector (Taticchi et al., 2015). The research paradigm itself is, at least under this term, quite new. Carter and Rogers (2008) claim to have taken the first steps applying sustainability to SCM. Since, an increasing interest in SSCM is documented by a rise in research contributions (Brandenburg and Rebs, 2015) and numerous special issues recently published in mainstream journals, such as Computers & Operations Research (Govindan and Cheng, 2015), European Management Journal (Choi et al., 2014), and Journal of Cleaner Production (Tseng et al., 2013). The paradigm of SSCM is directly deduced from the triple bottom line advising companies to consider the three pillars of sustainability: People, planet, and profit (Elkington, 2002). In many cases the boundaries between SSCM and adjacent paradigms, such as Green SCM (GSCM) and Corporate Social Responsibility (CSR) are blurry (Ahi and Searcy, 2013). Both can be seen as subsets of SSCM. While GSCM is located on the interface between lithosphere, technosphere and biosphere, CSR is interpreted as the social dimension of sustainable business practice (Gopalakrishnan et al., 2012).

Up to now, a consistent definition of SSCM is missing. Ahi and Searcy (2013) list twelve definitions used in contemporary publications. Close to the definition of sustainability, Ansari and Qureshi

(2015, p. 26) describe the SSCM paradigm as "maintaining a balance among social responsibility, environmental stewardship and economic viability along the entire supply chain, improving the long-term economic performance of an individual and the company and also meeting the customers' need competitively throughout the life cycle of goods and services". A broad understanding of SSCM is provided by Schaltegger and Burrit (2014) who define SSCM as the creation of economic value "with lower negative social and environmental impacts". Directly deduced from the triple bottom line, Kleindorfer et al. (2005) describe SSCM as "integrating profit, people and the planet into the culture, strategy, and operations of companies". Inspired by the definition of SCM (Mentzer et al., 2001) and the definitions given above, we interpret SSCM in a broad sense as planning, execution and control of corporate value creation processes by integrated consideration of economic, ecological and social aspects for the purpose of improving the long-term performance of an individual company and the supply chain as a whole. In line with this, the proposed framework targets a focal company that governs the supply chain, initiates sustainability activities, can force upstream companies to engage with sustainability initiatives and is decisive in designing the final product (Beske and Seuring, 2014).

2.2. Research gap

A meta-study of recent reviews suggest that the young and, in both research (Ashby et al., 2012) and industry practice (Varsei et al., 2014), still evolving field of SSCM can be described as fragmented (Gopalakrishnan et al., 2012) or – as Srivastava (2007, p. 68) coins it – "compartmentalized" with frequently discussed concepts such as management of closed-loop supply chains (Stindt and Sahamie, 2014), sustainable supply chain design (Eskandarpur et al., 2015), green procurement (Pimenta and Ball, 2015), socially responsible sourcing (Zorzini et al., 2015), green logistics (Dekker et al., 2012), sustainable manufacturing (Gahm et al., 2016), integrated chain management (Seuring and Müller, 2007), sustainable mining (Pimentel et al., 2016), performance measurement (Beske-Janssen et al., 2015) or information technology as an enabler for SSCM (Thöni and Tjoa, 2015). Attempts to map the field with a broader scope are confined to certain methodologies as quantitative modelling (Brandenburg et al., 2014; Brandenburg and Rebs, 2015; Seuring, 2013; Tang and Zhou, 2012), and empirical studies (Carter and Easton, 2011), or are limited to environmental issues (Gupta and Palsule-Desai, 2011; Srivastava, 2007). Although each of these articles extensively summarizes its epistemic field, they do not provide specific decision-support by integrating concepts and methods in a structured planning process. Addressing this challenge, a number of frameworks provide implementation principles and best practices. For instance, Seuring (2004) describes best practices for integrated chain management referring to companies in the textile industry. Tsoulfas and Pappis (2006) mention 25 principles for sustainability initiatives along the supply chain. To identify areas with significant potential, Lake et al. (2015) propose a four-step model. A reference process for generation of economic, ecological and social input parameters for subsequent mathematical analysis is given by Varsei et al. (2014). Schrettle et al. (2014) delineate how to select decision alternatives depending on the drivers that motivate SSCM. Based on observations at British Aerospace, Gopalakrishnan et al. (2012) develops a generic roadmap for actually implementing and steering sustainability initiatives. Unfortunately, these contributions on implementation principles and best practices focus on specific subsets of SSCM and mostly neglect methods for analysis, comparison and selection of adequate initiatives.

Summing up, a holistic synopsis of concepts and methods for

implementing sustainability in corporate supply chains is missing, although such a framework is strongly needed to transform the sustainability paradigm into the realm of industry practice. We set out to address this gap by developing a planning approach comprehensively supporting the development of sustainable supply chain strategies. Closest to this goal is Jaehn (2016) who develops a taxonomy of sustainable operations. He gives a broad overview of sustainability practices applicable by private corporations, public institutions and non-governmental organizations focusing on methods of operations research. Although this article shows certain overlaps, our article provides unique insights (i) by developing a planning approach tailored for private sector companies, (ii) by proposing methods exceeding those being part of operations research, and (iii) by eventually integrating these concepts and methods within a structured planning framework applicable to plan, steer and advance sustainability within corporate environments.

3. Methodological approach

This research is set out to synthesize concepts and methods provided by academia for the purpose of easing sustainability-oriented decision-making in corporate environments. To thoroughly answer the stated research questions in a valid and reliable manner, a structured research process is applied (Fig. 1) inspired by the procedures used in Gahm et al. (2016) as well as Khalid et al. (2015) extended by guidelines for content analysis in SCM (Seuring and Gold, 2012). In this sense, the research can be classified as grounded theory based on a qualitative content analysis of scientific literature (Bryman and Bell, 2011). For the purpose of advancement, verification and face validity, the research was discussed numerous times in master courses, at academic conferences as well as with knowledgeable colleagues as suggested in Bryman and Bell (2011).

The initial **conceptualization** of the epistemic object was informed by review articles within the field of SSCM (see Section 2.2). For the purpose of **material collection** we employed common principles for literature search (vom Brocke et al., 2009). As we like to reveal the concepts and methodologies applied to SSCM, we use a specific search string (“Sustainable Supply Chain”) to comprehensive databases, namely EBSCO and ScienceDirect. Within the databases we cover peer-reviewed journals published in English, except journals that are located in remote disciplines, like

medicine. A limitation with regard to the date of publication is not applied. Studies on food and agricultural supply chains (Ene et al., 2013), public waste management (Blengini and Garbarino, 2010), sustainable tourism (Font et al., 2008), green marketing and green accounting (Jasch, 2006, Schaltegger and Csutora, 2012) as well as studies taking a macro-economically perspective (Foran et al., 2005; Liao et al., 2015; Yu et al., 2010) are excluded. Editorials (Choi et al., 2014; Tseng et al., 2013; Govindan and Cheng, 2015) are also not considered. The **material evaluation** encompassed a total of 602 articles produced by the search query (EBSCO: 295; ScienceDirect: 307). The restriction to peer-reviewed journals reduced the number of articles to 394. After deletion of redundancies and a full-text review 133 articles are left for further analysis. The analytical categories used for **categorization and synthesis** are deductively identified from the literature basis. For the purpose of categorization, we distinguish between ‘concepts’ that represent general principles representing “building blocks of theory” (Bryman and Bell, 2011), such as Green Logistics, and ‘methods’ that help operationalizing a concept by means of systematic procedures by means of both data acquisition and decision-making. As some concepts and methods which are potentially valuable for SSCM are only discussed rudimentary within publications, a deeper understanding was generated by adding specific but randomly selected literature pertaining to these topics. Such **domain-specific enhancements** are especially required in the context of the social dimension (e.g., Social life cycle assessment) or specific domains of the environmental dimension (e.g., industrial symbiosis). The identified concepts and methods are mutually interrelated and positioned building a **SSCM framework** for corporate decision-making referring to generic **reference models for SC planning** presented in Adam (1997) and Klein and Scholl (2004).

4. Literature analysis

In this section we elaborate on the literature reviewed and analyzed within Step II. to IV. of the research process. We present a descriptive overview of the quantifiable statistics and derive performance indicators used in SSCM-oriented decision-making.

4.1. Frequency analysis and descriptive statistics

The analysis confirms the rising interest on SSCM (Fig. 2). In addition to the publication frequency, we extracted the

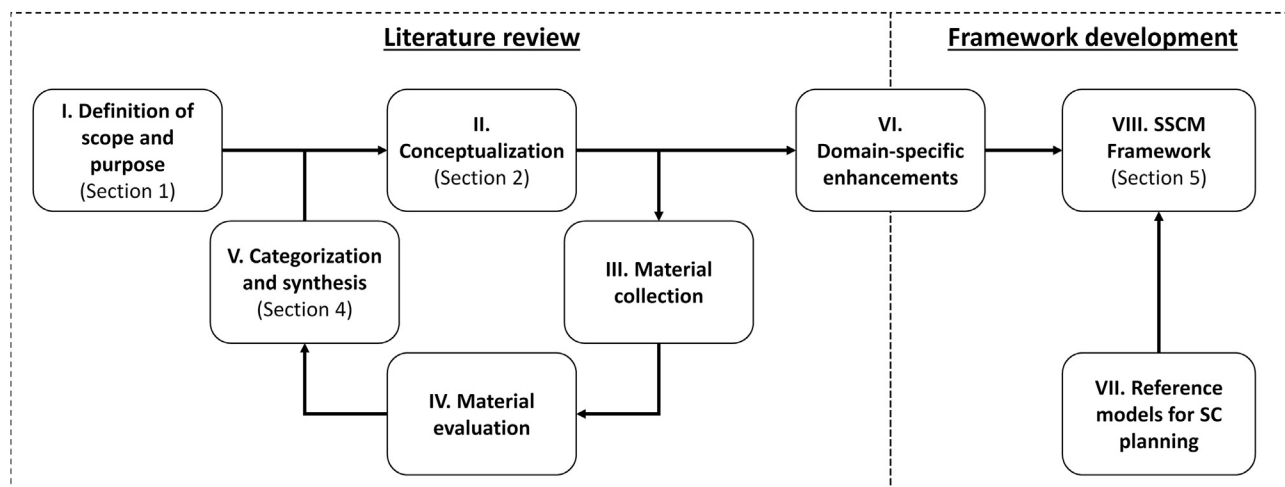


Fig. 1. Research process (inspired by Gahm et al., 2016; Khalid et al., 2015).

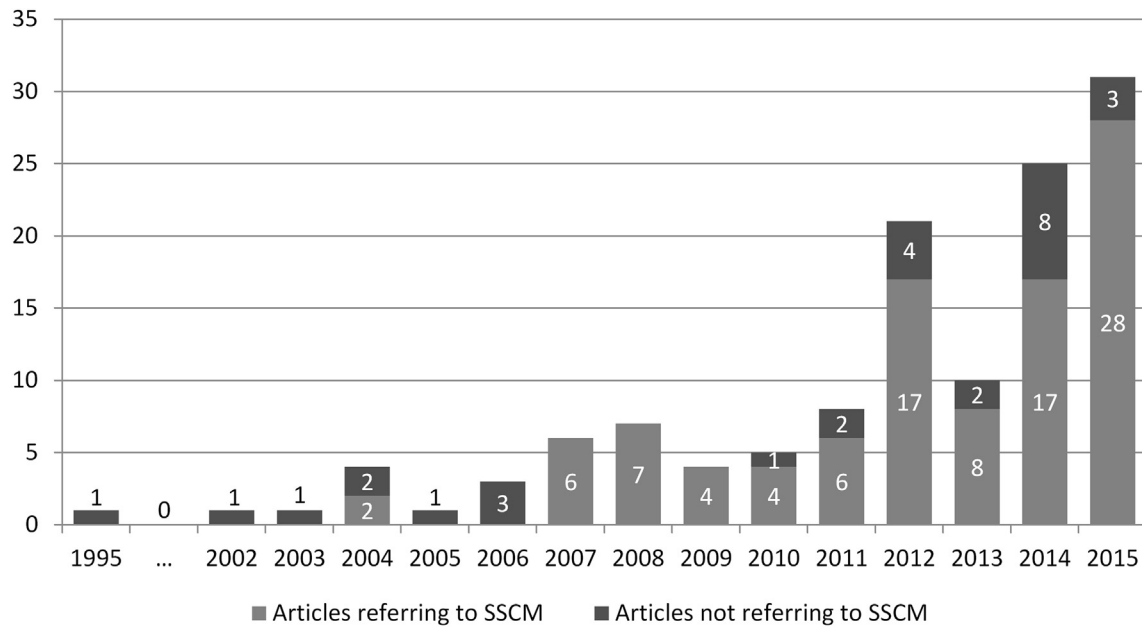


Fig. 2. Temporal distribution of articles (Five articles published in 2016 are not shown).

paradigmatic background of each study. Within the sample 77% of all studies are explicitly located in the context of SSCM. This means that the authors refer to SSCM for motivating their research or prominently introduce the paradigm. One tenth of all studies are embedded in GSCM (e.g., Bloemhof-Ruwaard et al., 1995; Maxwell and van der Vorst, 2003), while two are built on CSR (Amini and Bienstock, 2014; Nikolaou et al., 2013) and 13 studies do not mention any paradigm. In the temporal context, we see the SSCM paradigm gaining momentum by 2007. We observe a significant culmination of SSCM research in a limited number of journals: Journal of Cleaner Production (34 articles), International Journal of Production Economics (16), Supply Chain Management: An International Journal (12), International Journal of Production Research (10), International Journal of Physical Distribution and Logistics Management (8) and European Journal of Operational Research (5).

Table 1 shows the distribution of the publications across five research domains: Quantitative decision-support, qualitative decision-support, theory development, performance analysis, and articles summarizing the state-of-the-art. This categorization provides advantages for the upcoming framework development: First, it gives a hint concerning the orientation of the articles by means of being more academia-oriented or practitioner-oriented. Moreover, it helps identifying articles beneficial for decision-support in SSCM (for a detailed overview on the reviewed articles and the categorization see appendices). Table 1 also suggests that the vast majority of contributions at least address the economic and ecological dimension, with a significant fraction tackling the triple bottom

line as a whole. In the next section we provide an in-depth analysis of the sustainability indicators used. Here, we focus on quantitative decision-support as these studies need to explicitly define and operationalize these indicators.

4.2. Metrics and indicators within SSCM

Decision-support in SSCM is provided by means of normative mathematical models comprising optimization and simulation techniques, methods of multi-criteria decision making (MCDM) as well as analytical models. To provide adequate insights and to appropriately guide corporate SSCM initiatives, these models are required to cover multiple dimensions of sustainability (Matos and Hall, 2007). Table 2 gives an overview of the ecological and social issues tackled in quantitative research. The categories are adapted from the Sustainability Reporting Guidelines of the Global Reporting Initiative (GRI, 2012), which are well accepted by academia (Varsei et al., 2014; Eskandarpur et al., 2015) and are widely coherent with indicator sets proposed by other organizations and institutions, like the OECD, NIST, Guidelines of social accountability 8000, or the standards promoted by the International Labor Organization. Please be aware that the figures in Table 2 do not add up. On the one hand, several articles address more than a single issue. On the other hand, in some cases the authors do not refer to a specific issue but to an impact category in general.

Ecological initiatives pursue improvements of the input-output

Table 1
Distribution of articles across sustainability dimensions and research domains.

	Quantitative decision-support	Qualitative decision-support	Theory development	Performance analysis	State-of-the-art	Total
Economic	3	0	0	0	0	3
Ecological	1	5	4	0	0	10
Social	0	1	1	0	1	3
Economic & Ecological	18	6	12	6	5	49
Economic & Social	0	0	0	0	0	1
Ecological & Social	0	1	5	0	4	10
Triple bottom line	14	12	17	6	11	57
Total	36	25	39	12	21	133

Table 2
Environmental and Social Indicators applied to quantitative decision-support models.

Impact category		# of articles	Sustainability issue	# of articles
Ecological Dimension	Resource saving	21	Material intensity	8
			Recyclability of product	0
			Use of secondary input	8
			Energy efficiency	11
			Integration of renewable energy systems	0
	Mitigation of adverse environmental effects	31	Land use & Biodiversity	4
			Use/Release of toxic substances	7
			Reduction of emissions (e.g., CO ₂)	24
			Reduction of solid wastes	17
			Reduction of water consumption	6
Social Dimension	Labor practice and decent work	13	Environmental management system (e.g., ISO 14000)	3
			Employment effects and employee turnover	6
			Occupational health and safety	6
			Employee development	2
			Diversity among workforce	2
			Exploitative hiring policies	0
			Social standards (e.g., SA8000)	1
			Excessive working hours	0
	Human rights	4	Discrimination	0
			Freedom of Association	0
			Child and forced labor	0
	Society	7	Indigenous rights	0
			Impact on local community and economy	5
			Corruption	0
			Anti-competitive behavior	0
			Animal treatment	0
	Product responsibility	2	Socially undesirable products	0
			Customer Health and Safety	0

ratio by simultaneously reducing resource consumption and adverse impacts on the ecosystem, such as climate change, smog, acidification, or eutrophication (Bloemhof-Ruwaard et al., 1995, Tang and Zhou, 2012). Strategies for saving resources can be realized by designing less material intensive products (Tajbakhsh and Hassini, 2015) or products not containing depleted resources (Borchardt et al., 2011). Other authors concentrate on increasing the fraction of secondary input (Devika et al., 2014; Li, 2013). Energy efficiency, the leading issue for realizing resource savings, is either implemented by minimizing energy consumption during processing (Rager et al., 2015) or by designing products to be less energy intensive throughout usage (Kuo et al., 2010). Models that prefer energy sourced from renewable sources (e.g. scheduling of operations considering supply peaks from renewable energy systems) could not be identified. The effects of supply chain activities on biodiversity and land change, topics often quoted in qualitative literature (Beske-Janssen et al., 2015; Chardine-Baumann and Botta-Genoulaz, 2014), are widely neglected in modelling. Being unaware of effects concerning land use change and biodiversity can bear significant economic risks as experienced by sectors publicly criticized for deforestation connected to production of leather and palm oil (Vurro et al., 2009; Wolf, 2014). Especially in the mining sector, the impact on land and land rehabilitation upon closure of sites represents a decisive issue (Pimentel et al., 2016). Parameters representing adverse eco-effects mostly emphasize greenhouse gases emissions, either referring to total emissions (Chaabane et al., 2012), to the carbon footprint (Jakhar, 2015) or to emission intensity (Boukherroub et al., 2015). Further objectives comprise the reduction of wastes and unwanted by-products as well as water consumption (Geldermann et al., 2007) or the utilization of toxic substances (Kuo et al., 2010). Despite certain problems concerning the determination of precise ecological impacts, these indicators are mostly measurable at a cardinal level easing the integration into quantitative decision-making models.

The social dimension is measured nominally in many cases (Schaltegger and Burrit, 2014). In consequence, the incorporation of

social indicators is less elaborated. In many cases, singular indicators, such as employment (Jakhar, 2015), employment stability (Boukherroub et al., 2015), injury rates (Bouchery et al., 2012) as well as job creation in less developed regions (Mota et al., 2015) are taken into account. Impact on local communities can be operationalized in terms of donations, community projects or cultural services provided by the focal company (Govindan et al., 2013; Su et al., 2016). Table 2 shows a number of issues not addressed at all. Some authors argue that an improvement of the environmental performance is an implicit contribution to the social dimension (Ashby et al., 2012; Schrettle et al., 2014). The approach presented by Tseng and Hung (2014) can be regarded as an application of this understanding calculating a social cost rate for damages to human beings induced by carbon emissions.

In addition to the insights specific to each sustainability dimension, we observe common patterns between them. Measures of sustainability commonly refer to top level concepts, like human rights (Kuo et al., 2010), or apply rather abstract indicators, like CSR performance (Hsueh, 2015) or sustainability degree (Li and Li, 2016), staying vague about the practical meaning (Bouchery et al., 2012). Moreover, quasi-economic objectives are used as ecological goals, e.g. minimization of perishable left-overs (Choi and Chiu, 2012; Soysal et al., 2015) and eco-design costs (Azadi et al., 2015), or as social objectives respectively, e.g., corporate reputation and customer satisfaction (Tajbakhsh and Hassini, 2015). Environmental (e.g., ISO 14000, AA1000) and social certifications (e.g., SA8000, OHSAS, 18001, ISO 26000) or codes of conduct are not directly used as metrics but are frequently used to signal a desired behavior or are seen as minimum criteria (Morali and Searcy, 2013).

The content analysis reveals that the environmental and, a fortiori, the social dimension is often only rudimentary treated. Concluding, shortcomings pertaining to the comprehensive integration of sustainability factors within quantitative decision-support models outlast, leading to the fact that comprehensive transition of sustainability considerations into quantitative decision-support is rare. We believe that the structured approach

synthesizing relevant concepts and methods within the proposed framework helps to overcome these shortcomings and, thus, contributes to the development of more realistic and holistic decision-support models.

5. A framework for SSCM decision-making

In this section, we elaborate on the framework for guiding decision-processes in SSCM (Fig. 3). The central contribution is the structured integration of paradigms, concepts, and methods utilizable in the pursuit of ecological and social improvements along the value chain. The framework builds on the generic phases of planning comprising three consecutive steps (Adam, 1997; Klein and Scholl, 2004) and is inspired by the transposition framework presented by Boukherroub et al. (2015). Building on these generic reference processes for decision-making in supply chain management, we guarantee the applicability of the framework to various SSCM-oriented planning problems in corporate environments.

- Phase I: In the first phase the characteristics of the problem are defined and a qualitative goal is derived (e.g., reducing carbon emissions). Executives may select one of the concepts proposed in section 5.1 that (i) fit the strategic goals and business concept

of the focal company and (ii) address the sustainability goal at hand. The qualitative definition of goals, the description of the problem, and the information requirements can be narrowed by concept selection.

- Phase II: The second phase is dedicated to the identification and assessment of potential decision alternatives, including investments, process changes, and product changes. Methods capable to capture relevant parameters (see section 4.2) are proposed in section 5.2. In this way, the quantitative parameters for each alternative and the decision variables are determined. Based on decisive data, a quantified goal is formulated.
- Phase III: Depending on the size and complexity of the problem a decision-support method is applied and fed with the parameters determined in the preceding phase. This may happen using simple decision rules or sophisticated methods such as the ones described in section 5.3. A crucial issue at this point is the handling of competing objectives. Eventually, the selected decision alternative is carried out either by means of investments, process changes or product changes.

Before describing the tools for each phase in detail, we like to clarify basic terms, which are defined by consultation of the Oxford Dictionary (2015) and Merriam Webster (2015). A 'paradigm' encompasses selected concepts that reflect a world view determining

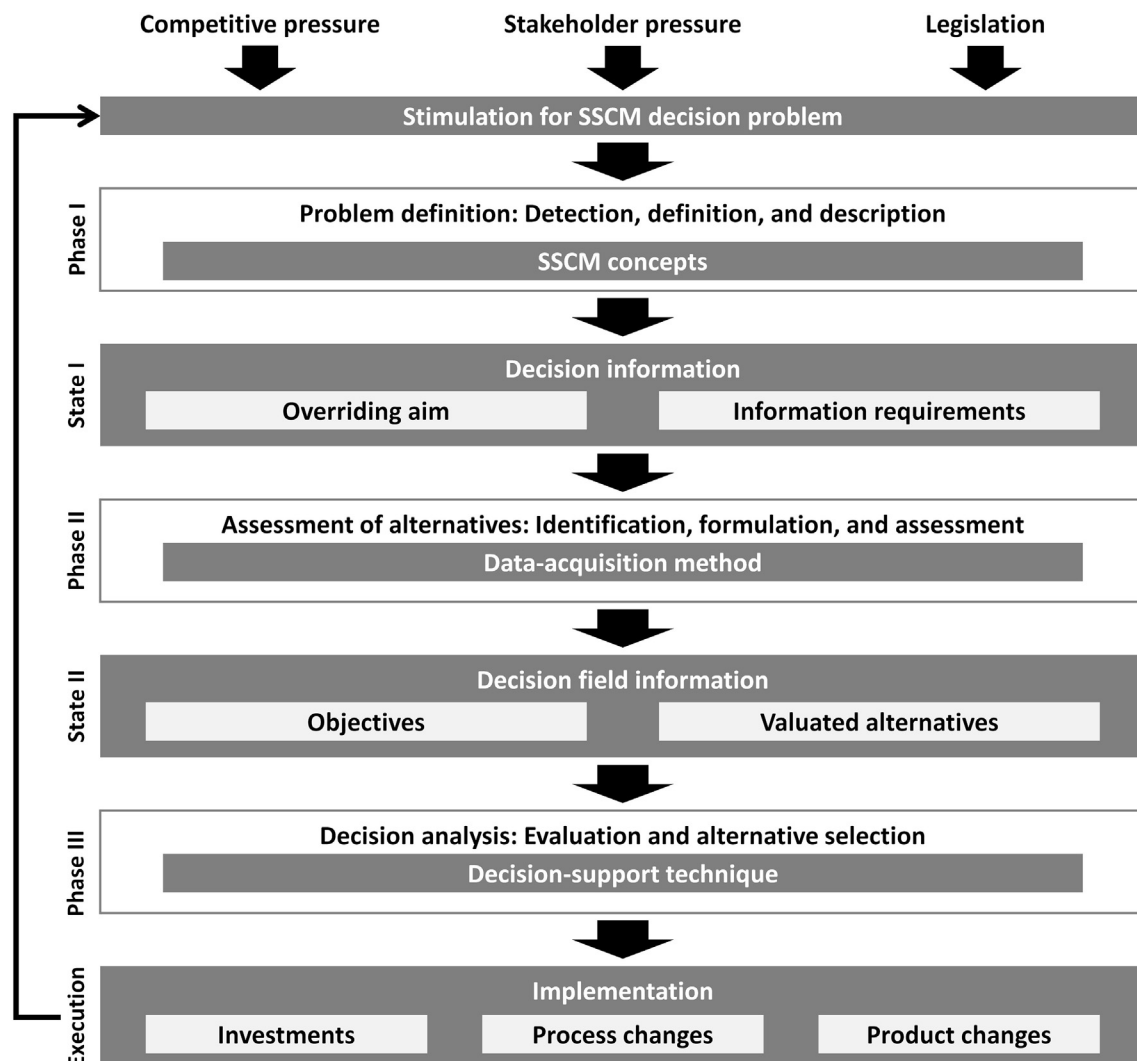


Fig. 3. A framework for SSCM decision-making.

patterns about the way something should be thought about. SSCM as well as its subsets GSCM and CSR are paradigms as they changed the view business practice is perceived by stakeholders. To move from paradigm to something more concrete, 'concepts' represent general principles that are connected to something abstract. In this sense, Green Logistics, Sustainable Procurement and alike are classified as concepts. 'Methods' help to operationalize a concept by means of systematic procedures.

5.1. Concepts of SSCM

In total we identify nine concepts for improving sustainability along the value chain. Each of the concepts is connected to value chain processes, which are defined following the Supply Chain Planning Matrix (Fleischmann et al., 2008), the SCOR model (Ntate et al., 2015), and the Reverse Supply Chain Planning Matrix (Nuss et al., 2015) covering all phases along which SSCM practices can be implemented: Raw Material Mining, Sourcing & Supply Chain Coordination, Production, Distribution, Use, and Return (Fig. 4). We introduce each of the concepts in the following.

Sustainable Product Design: Traditionally, companies' strive for developing products of a certain quality with least costs. Nowadays, the environmental and social performance of products over the whole lifecycle has to be considered. From the ecological perspective, the product design should result in products that are characterized by decreased material intensity, less toxic inputs, biodegradability, durability, ease of recovery and low energy consumption over lifetime (Leigh and Li, 2015). A majority of impacts generated over a product's life cycle is determined at the development stage with options for material input, manufacturing processes, impacts during use, and end-of-life treatment being set to a great extent (Seuring, 2004). Addressing the variety of issues, we differentiate four broad categories of Sustainable Product Design: First, product design may target a reduction of the quantity or quality of input materials build into the product by decreasing the material intensity, increasing the fraction of secondary inputs or limiting the use of toxic substances (Pimenta and Ball, 2015). Such design objectives result in resource conservation and less impact on both the environment and human beings ('Design for Material Efficiency') as mining processes are a major source of emissions and of damages to landscape and biodiversity (Hassini et al., 2012).

Moreover, certain sectors of the mining industry are notorious for social risks, like forced and child labor. Secondly, considerations may be given to the impacts during production ('Design for Sustainable Production'), such as reducing the amount of production wastes, including by-products, water consumption and auxiliary materials. From the social perspective, the product design may influence the ease of handling during production as well as the risk of occupational injuries induced by certain assembly steps or by noxious materials impacting the workforce. Third, focusing on the use phase ('Design for Sustainable Usage'), consumer risks associated with the product are to be considered. In a broad understanding, decisions may also comprise product portfolio issues leading to a phase-out of products that are perceived as socially undesirable. Environmental factors within the use phase mostly pertain to energy efficiency and length of product life-cycles (Handfield et al., 2005). A fourth category deals with the influence of product design on recoverability ('Design for Recovery'), e.g., modular assembly opens up a broad set of recovery options, ranging from recycling to remanufacturing (Bansal, 2005; Veleva and Ellenbecker, 2001).

Sustainable Supply Chain Design: Supply chain design is a cross-cutting planning problem embracing all value chain processes of the focal company with interfaces to suppliers and customers representing sources and sinks of material flows. As being highly strategic, decisions on the supply chain design delimit the degrees of freedom within almost every other planning problem of the company's operations (Nuss et al., 2015). Common decisions deal with the localization of production sites and distribution facilities framed by the geographical dispersion of supply sources and customer demand. Apart from the location problem, decisions on site-specific capacities, technology choices and flows between the entities are often interwoven. The environmental factors typically considered within this type of problem deal with emissions. The impact of land development on the biosphere, which would be of special interest in 'greenfield' problems, is widely ignored. The social lens is even less developed. Some authors merely give advantage for creating jobs in less developed regions.

Sustainable Procurement: Irresponsible behavior by one member affects the image of the supply chain as a whole by integrating environmental and social aspects into sourcing decisions (Mitra and Datta, 2014; Zorzini et al., 2015). Sustainability efforts of

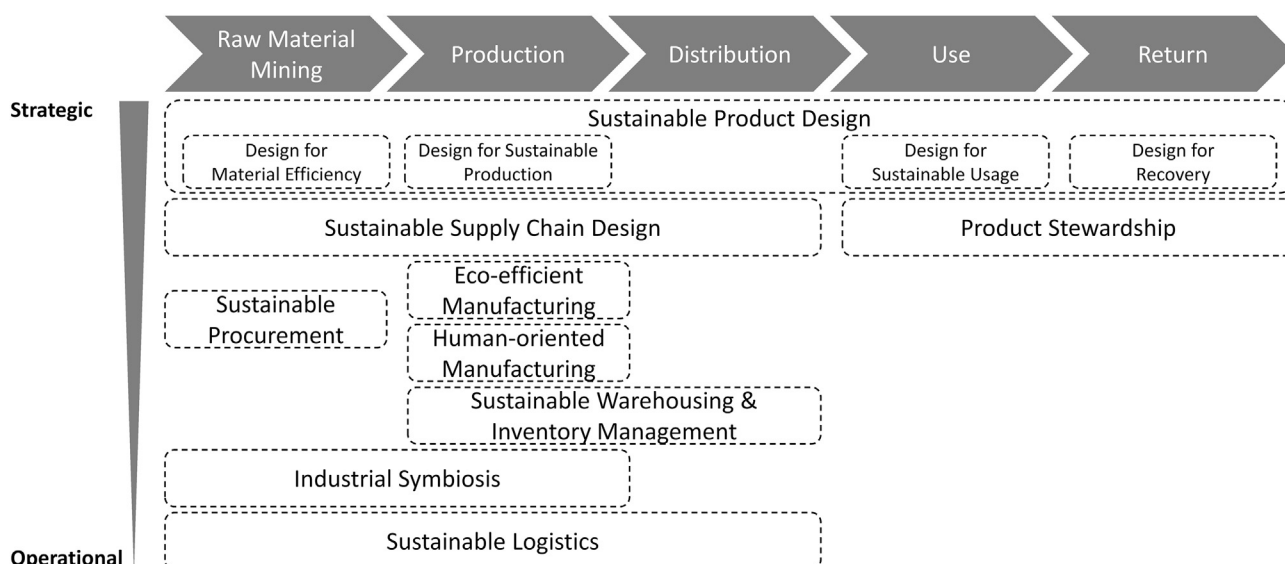


Fig. 4. SSCM concepts along the value chain.

the focal company are passed to upstream supply chain actors by exertion of purchasing power. Sustainable Procurement can be realized by setting requirements concerning the sustainability impact of the sourced inputs (e.g., conflict-free materials) or by insisting on suppliers' adherence to ecological and social standards within their processes. According to supplier assessments and the subsequent selection comprise auditing and monitoring of suppliers encompassing site visits, in-depth analysis of suppliers' business model and supplier surveys (Seuring, 2013). Such practices generate reliable data, but are costly at the same time. Instead, many corporations rely on external certifications, verifiable processes such as environmental management systems in line with ISO 14001, or supplier's commitment to a code of conduct (Turker and Altuntas, 2014). In contrast to supplier selection, collaborative approaches for supplier development imply intensive relationship management involving supplier education programs and inter-organizational working groups (Leigh and Li, 2015; Leppelt et al., 2013). Such strategic collaboration is supposed to be superior concerning the success of sustainability initiatives compared to approaches that purely rely on supplier assessments (Sancha et al., 2016; Vachon and Klassen, 2006). A number of sustainability indicators is specific to procurement comprising factors like the fraction of suppliers that are either located nearby, are under control of minorities (e.g. indigenous people), or are located in developing countries in order to strengthen the regional economy (Hassini et al., 2012; Pagell and Wu, 2009). While most studies target first tier suppliers, proactive firms are supposed to implement direct (interorganizational collaboration) or indirect (e.g., list of prohibited substances, certification requirements) governance mechanisms pushing sustainable behavior further upstream (Tachizawa and Wong, 2014).

Sustainable Manufacturing: Sustainable Manufacturing is considered a cornerstone of sustainable development (Garetti and Taisch, 2012). Within this analysis, we differentiate between two concepts: Eco-efficient Manufacturing and Human-oriented Manufacturing.

Eco-efficient Manufacturing encompasses technological and organizational measures that result in ecologically benign production processes that are less-polluting, energy efficient, and emit a minimum of unwanted substances exceeding traditional make-span-, throughput-, or capacity-oriented objectives (Tang and Zhou, 2012). On a strategic level, investing in innovative eco-friendly equipment is seen as a major driving force for advancing ecological production (Schrettle et al., 2014). For existing machinery, re-configurations can induce environmental benefits. Majozzi (2005) shows that alterations of layout design can generate situations where profit is improved and resource consumption is reduced at the same time. Another option for cleaning the production process is to equip production plants with state-of-the-art end-of-pipe technologies (Garetti and Taisch, 2012). Organizational measures on the tactical and operational level can substantially improve the environmental performance without necessitating high investments. Rules of thumb regulating the shut-down of idle machinery (Pach et al., 2014) or simple measures like eliminating leakages, optimizing drives, saving of solvents and lubricants, and recovery of by-products can also contribute to greening (Duflou et al., 2012; Haapala et al., 2013). More sophisticated approaches encompass optimization of production schedules, cutting processes, and maintenance cycles. Scheduling approaches are mostly input-oriented with energy-efficient production planning representing a growing field of research (Gahm et al., 2016), while output, e.g. wastewater reduction, is less focused (Giret et al., 2015). The breadth of energy saving strategies in production environments is compiled in Abdelaziz et al. (2011). Cutting optimization contributes to the reduction of wastes and by-products (Dyckhoff,

1990; Wäscher et al., 2007). Although being well elaborated, cutting problems have not been associated with SSCM so far. Improved maintenance cycles benefit sustainability by extending the equipment's lifetime, and by reducing the fraction of rejected products (Garetti and Taisch, 2012). Potentials neglected in Eco-efficient Manufacturing pertain to the integration of renewable energy systems, e.g., by developing production plans in line with supply patterns from renewables or by analyzing the viability of energy storages build into production processes.

Human-oriented Manufacturing mainly deals with measures for employee safety and working conditions. This is in contrast to strategic social issues (e.g., wage level, conforming human rights), which are focused else. Sarshar et al. (2016) and Bautista et al. (2016) give an overview of planning issues that impact the risk of major accidents and propose countermeasures. In contrast to these acute incidents, long-term and more subtle effects on the workers well-being, like musculoskeletal disorders caused by repetitive work, are barely discussed. For instance, ergonomic considerations of the working place and material handling are missing in SSCM models (Pagell and Shevchenko, 2014). Fabbro and Santarossa (2016) may help to exploit this topic. Moreover, contemporary production planning assumes employees to be interchangeable, emotionless and deterministic ignoring the workforces' behavior and their individual skills (Grosse et al., 2015). The product recovery operations of a major electronics manufacturer, which we could observe within action research, illustrate the importance of these aspects. In line with the common heuristics for production planning, the manually conducted disassembly tasks were organized as batches of similar products. This organization ended up in decreased productivity. Eventually, employees were given the freedom to decide on the sequence of tasks on their own which resulted in increased motivation, a higher level of satisfaction and increased productivity. The motivational effect of having control over a task is long known within research on human resources management (Hackman, 1978) but did not trickle into SCM. Likewise, issues pertaining to employee training, corporate culture and reward structures are missing (Jabbour and de Sousa Jabbour, 2016). Another topic of interest from a social perspective is the adaption of planning models considering the specific requirements of disabled people. Decision-support in this area would help to remove executives' concerns and may facilitate the inclusion of minorities.

Sustainable Warehousing & Inventory Management: Problems of warehousing traditionally deal with equipment choices within the warehouse as well as optimization of order quantities and order frequency. High energy demand and according emissions for storing products that need temperature control imply significant impacts on the environment (Dekker et al., 2012). Intelligent warehousing may also contribute to waste prevention by minimizing the amount of perishable left-overs. The social issues tackled in sustainable warehousing correspond to those described in human-oriented manufacturing. Material handling is a main task in warehousing operations and should be conducted in a way that is not physically and mentally straining. Grosse et al. (2015) names human factors within planning of order picking referring to factors like size, weight, volume, and location of the product. Multi-criterial examinations of inventory management having not been existent half a decade ago (Hassini et al., 2012) are now evolving mostly as extension of the economic order quantity (EOQ) model (Battini et al., 2014).

Industrial symbiosis: Manufacturing concepts focus on production processes of a focal company, whereas Industrial Symbiosis deals with improvements realizable by cooperation of facilities under control by different companies. It promotes resource efficiency by establishing eco-industrial parks that are characterized

by a symbiotic collaboration of the involved actors based on the exchange of resources. Up to now, Industrial Symbiosis is widely neglected by mainstream SSCM research and the overall literature on the interface between both is underdeveloped. This is unfortunate as Industrial Symbiosis may yield significant environmental and economic gains, and promising fields for application are already identified (Leigh and Li, 2015). Companies striving for more efficient management of material and energy flows are likely to benefit by realizing potentials for reduction of wastes and unwanted by-products (Winkler, 2011). Zhu and Cote (2004) provide an in-depth description of an eco-industrial park highlighting levers and barriers faced during run-up and operations.

Product Stewardship: The research field pertaining to end-of-life management evolved from disposal management to circular approaches. Product Stewardship aims on establishing product recovery systems and, thus, is strongly intertwined with closed-loop supply chain management which is perceived as “sustainable almost by definition” (Quariguasi Frota Neto et al., 2010, p. 4463). It is supposed to reduce waste, mitigate the effects of toxic substances, and save resources initially invested into the product (Ashby et al., 2012). Socially, closed-loop supply chains create jobs for less-skilled workers on a regional level as recovery processes are barely automated. Circular economy concepts can be realized by recovery of internal by-products or by collecting products from customers after use (Stindt and Sahamie, 2014). The management of product backflows induces a totally new set of managerial problems, which are mainly caused by uncertainties regarding timing, quantity and quality of returns. Nuss et al. (2015) describe these challenges along the processes of product returns management, reprocessing operations, and remarketing. The range of challenges is rather wide including backflow forecasting, infrastructure decisions, disassembly planning and redistribution planning, but also comprises issues such as hazard management (Diabat et al., 2014) and save disposal of harmful substances (Geldermann et al., 2007).

Sustainable Logistics: Waste from means of transportation as well as emissions generated through transportation are the major source of environmental impact in logistics (Garetti and Taisch, 2012). Within the struggle for sustainability, corporate executives perceive savings in these areas as a powerful lever (Ageron et al., 2012). Sustainable Logistics is often limited to green issues, which are reviewed in detail by Dekker et al. (2012): Practices such as reducing the frequency and distance of transports, optimization of routing, efficient container loads, and multimodal transport are named. Typical social issues within logistics deal with injuries and deaths caused by accidents, fair payment of truck drivers, avoidance of congestions, noise reduction, as well as ergonomic handling of goods (Nikolaou et al., 2013). Even less assessed is the impact of transportation on land use and biodiversity, although both is strongly influenced by the infrastructure that is required, be it roads, waterways, harbors, or airports.

5.2. Data acquisition for ecological and social assessment

In line with Phase II, methods utilizable for acquisition of the environmental and social data (see Section 4.2) are proposed (Fig. 5). Material flow analysis (MFA) provides a structured approach for depiction of inputs and outputs along value chain processes. Mechanisms for aggregation and rules for assessing the severity of impacts are described in footprinting approaches, life cycle assessments (LCA) and criticality assessments enabling ordinal comparisons of sustainability performance. In the following we introduce these methods and highlight some contributions that exemplify their application.

Material flow analysis visualizes input-output flows at every stage of a process. Value stream mapping may serve similar purposes. These tools produce sound knowledge and graphical depiction of material flows and waste flows identifying efficiency losses (Seuring and Müller, 2007). Apart from solid substances, this methodology can be used for analyzing emission sources and

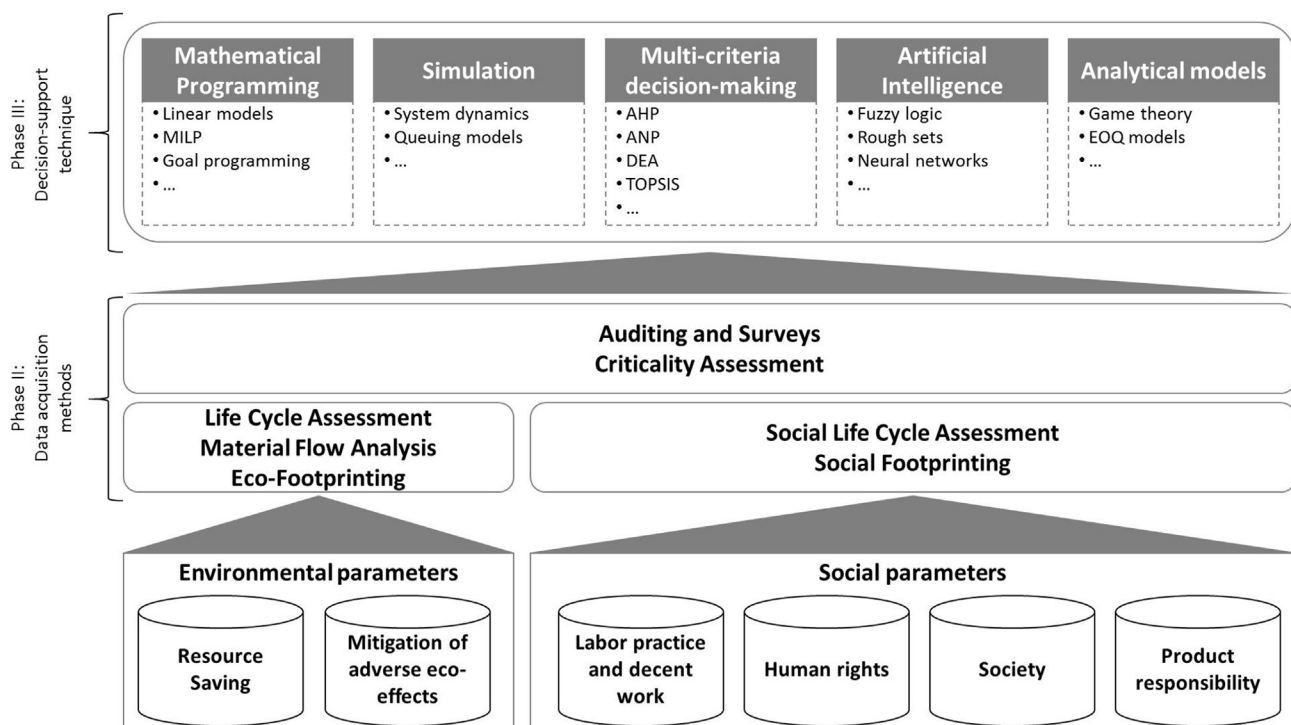


Fig. 5. Tools for assessing SSCM alternatives and decision-making.

according reduction potentials (Ball et al., 2009). Furthermore, it is useful for comparison of ex-ante and ex-post states. MFA is regarded as a prerequisite for improvements concerning waste generation and emissions, especially in manufacturing (Giret et al., 2015). For this purpose, Smith and Ball (2012) establish guidelines for utilization of flow modelling. Rybicka et al. (2015) demonstrate the benefits of MFA within a production system for composite materials revealing potentials for by-product recovery. More ambitiously, Ball et al. (2009) use MFA to establish zero waste and zero emission manufacturing practices within a system consisting of energy utilities and manufacturing processes.

Ecological footprints compare the extent of human interaction with the capacities of nature. Human activities are perceived as sustainable when natural limits are not exceeded (Wiedmann and Barrett, 2010). Coming from a macroeconomic perspective, footprinting has been adapted to supply chain activities (Minx et al., 2009). The methodology translates ecological impacts into an aggregated single score, such as the total land area required to support the company's resource consumption and disposal (Barrett and Scott, 2001). Indeed, footprinting approaches are scarcely chosen and are limited to carbon footprints, although various inputs and outputs can be analyzed. Jakhar (2015) and Bouchery et al. (2012) consider the carbon footprint to select suppliers and to evaluate order processing activities. Reviewing the literature it is striking that the term 'carbon footprint' is often falsely used representing the total emissions of CO₂ associated with a process or a product. A more rigor distinction is necessary. In analogy to the ecological footprint, the **social footprint** measures social sustainability based on the needs of individuals. It gives the capability to quantitatively depict the social bottom-line of an organization using social indicators which are compartmentalized into four broad categories: Social capital, Human capital, Natural capital, and Constructed capital. The performance in each category is normalized by the so called people feet, which represents the fraction of human resources under supervision of the company. The social footprint itself is defined as the quotient from the company's performance within a selected sustainability category and a given target value expressing the carrying capacity of a capital category (McElroy et al., 2008). In contrast to the ecological footprint, a footprint value of one or above suggests social sustainability (McElroy, 2007). An overview covering both ecological footprinting approaches and various social footprints is provided by Cucek et al. (2012).

Environmental Life Cycle Assessments (ELCAs) exceed both the company's boundaries and the focus on material flows. Depending on the perspective and the epistemic object, LCA can serve as a standalone decision-support tool (Yung et al., 2012) or as a supportive tool for subsequent analysis. ELCA is a central tool for ecological evaluation of processes and products alike measuring environmental as well as health impacts at all stages of the value chain (Beske-Janssen et al., 2015; Rubin et al., 2014). Although still evolving, ELCA is scientifically accepted as being a reliable tool for environmental evaluation (Mota et al., 2015; Winkler, 2011). The reference procedure for conducting an ELCA is standardized by ISO14040. In total, ELCAs show a number of advantages: First, it covers the full breadth of environmental effects associated with business processes, ranging from CO₂ emissions to resource depletion. Second, ELCAs use statistical data which are extractable from databases, like EcoInvent. This significantly eases application and reduces the costs of primary data collection. A detailed overview of databases and software tools supporting the application of LCAs to manufacturing environments is provided by Mani et al. (2014). A drawback of LCAs is the variety of life cycle impact assessment methods that differ with regard to normalization assumptions and weighting factors (Carvalho et al., 2014). As a consequence, the results of a specific ELCA are not ultimately

accepted (Lake et al., 2015). Nevertheless, ELCAs are the predominant methodology for evaluating the ecological performance of a product satisfying the requirement of life-cycle thinking in Sustainable Product Design (Kjaerheim, 2005). Haapala et al. (2013) highlight the benefits of LCAs used within decision-making on Eco-oriented Manufacturing. In this context, ELCAs are utilizable in investment decisions revealing the equipment's environmental impact over its lifecycle, whereas, process-based ELCAs are applied to compare organizational alternatives. Evaluations of the environmental impact of Sustainable Supply Chain Design choices are also conducted by means of ELCA (Pagell and Shevchenko, 2014). A broader overview of respective applications is given by Eskandarpur et al., 2015. Further examples can be found in Kostin et al. (2015) who utilize the Eco-Indicator 99 to feed their models as well as Mota et al. (2015) who refer to ReCiPe for designing a sustainable network.

An outstanding attempt to quantify social aspects is presented by Pishvae et al. (2014). They calculate social parameters using **Social LCA** (SLCAs) following the 'Guidelines for Social Life Cycle Assessment of Products'. Emerging from ELCAs, the research on SLCAs is in its infancy with first publications a decade ago (Dreyer et al., 2006). Nevertheless, it is promising to determine the social performance associated with an organization, a product or a process (Pimentel et al., 2016). SLCAs may serve two main purposes: First, the structured methodology forces to define a scope for the social analysis, a task that is often left behind in social assessments (Eskandarpur et al., 2015). Second, it generates a manageable amount of parameters condensing numerous indicators leading to computable data. Interested readers refer to Hutchins and Sutherland (2008), Jorgensen et al. (2008), and Wu et al. (2014). Former provide an example on how to apply SLCA to a supply chain, the latter two introduce the methodology and review recent developments. An extensive guideline for product-based SLCA is provided by the United Nations Environment Programme (2009).

Criticality assessments are originally developed for analyzing the long-term availability of functional materials based on biophysical, technical, economic, and social factors (Bensch et al., 2014). Several aggregation mechanisms, indicator sets and data sources are proposed in literature (Achzet and Helbig, 2013; Bensch et al., 2015). Formerly, criticality concepts concentrate on the economic and the ecological dimension. Lately, a growing importance is given to the social impact of raw material extraction referring to indicators like human toxicity, child labor, and occupational health (Reller, 2011). Criticality assessment enables to capture impacts in a structured manner and proposes indicators as well as aggregation mechanisms. In particular, criticality assessment can bear benefits in Sustainable Product Design, especially concerning 'Design for Material Efficiency', and Sustainable Procurement by revealing the total costs and risks of materials. Giving transparency to the economic, ecological, and social impact of design alternatives, Bensch et al. (2015) assess design alternatives of circuit boards.

Auditing and Surveys aim on verifying the implementation of systematic organizational measures that allow reactive approaches for advancing sustainability within an organization. Auditing procedures often refer to standards such as ISO 14001. Auditing and surveys are commonly applied for assessing alternative suppliers in Sustainable Procurement. For these purposes Keating et al. (2008), Koplin et al. (2007) as well as Graafland (2002) provide in-depth portrays of best practices observed at a global electronics company and a German textile manufacturer respectively, describing reference processes, standard questionnaires and auditing procedures. From an operations research perspective auditing and surveys mostly translate to binary parameters reflecting the implementation of certain procedures.

Summing up, depending on the scope of the problem and the

focused SSCM concept the described methods facilitate the acquisition of appropriate data sets for potential decision alternatives. In this way, the fundamental data subsequently used as parameters and decision variables in normative decision-support models are accessible.

5.3. Methods supporting quantitative decision-making in SSCM

Methods supportive in SSCM decision-making have to be able to converge multiple criteria and to generate an ordinal order of the multi-attribute alternatives. Therefore, the methods have to handle conflicting objectives and have to reflect the decision-makers' preferences. Following the classification of [Brandenburg et al. \(2014\)](#), we mainly identify five types of modelling methods applied as quantitative decision-support tools: Multi-criteria decision-making (MCDM) approaches, artificial intelligence (AI), multi-objective programming methods (MOPM), simulation techniques, and analytical models. These methods mainly vary concerning the number of comparable alternatives, the required level of parameter homogeneity and the level of uncertainty.

Methods of multi-criteria decision making, including AHP, ANP, TOPSIS, and DEA, and **Artificial Intelligence**, such as artificial neural networks, fuzzy logic and rough sets, enable comparison of both qualitative and quantitative information in line with decision-makers' preferences and can deal with incomplete information ([Madu et al., 2002](#)). Although powerful for decision-support, these methods are only applicable to a limited set of discrete decision alternatives. MCDM and AI methods are mostly applied to issues of Sustainable Procurement. [Govindan et al. \(2013\)](#) present a fuzzy TOPSIS approach enabling the integration of qualitative information, including those describing the social dimension, for holistic evaluation of suppliers. [Kuo et al. \(2010\)](#) utilizes ANP to determine weights for balancing indicators in supplier selection combined with ANN. Using DEA, [Azadi et al. \(2015\)](#) provide a generic supplier selection model taking into account work safety and labor health. A similar proceeding can be adopted for comparing design alternatives for products and processes. Addressing the difficulties concerning the integration of social aspect, [Neumüller et al. \(2015\)](#) propose an ANP based approach for locating distribution centers covering sustainability impacts during operation, build-up and phase out of a facility. Particularly tackling the issue of uncertain and partially unknown data, a hybrid methodology based on grey theory and DEMATEL is applied by [Su et al. \(2016\)](#).

Multi-objective programming and **Simulation** are applied in situations where the number of decision alternatives is virtually infinite and the specific results of each solution are unknown making an analytical enumeration of alternatives impossible. As a downside, these techniques can hardly deal with unknown measures, non-cardinal indicators and 'soft' facts. While optimization assumes deterministic behavior of system parameters, simulation takes stochastic processes into account. A particular challenge in mathematical programming is how to capture and balance trade-offs between multiple objectives. In contrast to MCDM, mathematical programming approaches are not explicitly designed to simultaneously integrate multiple objectives. Acknowledging that variations and hybrid construction of approaches for integration of non-economic information exist, we distinguish three approaches: Monetization of ecological and social effects, epsilon-constraints, and utility values. All methods strive for selection of pareto-optimal, non-inferior solutions for the problem at hand.

Monetization of sustainability indicators is popular for environmental parameters such as costs induced by carbon emissions ([Tseng and Hung, 2014](#); [Diabat and Al-Salem, 2015](#)), fuel consumption ([Soysal et al., 2015](#)), or waste disposal ([Battini et al., 2014](#); [Soysal et al., 2015](#)). This conversion into monetary terms can easily

be accomplished when ecological and social effects are directly attributable to operational costs, such as disposal costs or – if operations are covered by an emission trading scheme – emission costs. Apart from the fact that such market prices are hardly obtainable for other indicators of sustainability (e.g., injuries, deaths, biodiversity) it is questionable if the real or fictional market price adequately reflects the 'natural' and 'social' costs. An attempt to monetize the social dimension is presented [Azadi et al. \(2015\)](#) who calculate the costs associated with safety and labor health.

Approaches implementing epsilon-constraints optimize a primary objective considering further objectives as constraints. Optimizing the economic performance, [Mota et al. \(2015\)](#) choose epsilon-constraints to integrate ecological and social objectives of the Sustainable Supply Chain Design problem. In a similar problem, [Kostin et al. \(2015\)](#) also propose epsilon-constraint methodology to reduce the complexity of a MOLP applied to supply chains of bio-ethanol and hydrogen production. In addition, the authors show that certain objectives are redundant and can be omitted without compromising the solution quality. Less frequent is the utilization of non-economic goals as primary objective. [Rager et al. \(2015\)](#) present an optimized production schedule levelling the energy demand. With this organizational measure, carbon emissions within a textile production process are decreased by 20% without compromising economic objectives. In a similar setting, [Denz \(2015\)](#) offers an optimization model for planning an energy supply system that operates close to its optimum operating point increasing energy efficiency based on existing production plans.

The utility value approach weights each objective converting the resulting value of the multi-objective problem into a single scalar. This approach can be found for all kinds of SSCM concepts, but especially in Sustainable Supply Chain Design problems. For instance, [Boukherroub et al. \(2015\)](#) balance the financial performance with emissions and the effects on local employment within a value chain for lumber products. Balancing the objectives pertaining to material consumption, emissions, and wastes, [Geldermann et al. \(2007\)](#) implement equal weight coefficients. [Pishvae et al. \(2014\)](#) propose a fuzzy linear membership function enabling the decision-maker to adapt the importance of each objective in accordance to individual preferences. Out of the sparse contributions targeting Human-oriented Manufacturing, [Yousif et al. \(2011\)](#) uses a weighted linear program to incorporate ergonomic factors into disassembly scheduling.

Other authors circumvent the problems of balancing the objectives by implementing a combination of decision-making methods. [Validi et al. \(2014\)](#) compares the feasible solutions of a bi-objective mixed integer program using TOPSIS to reflect the decision-makers' opinion. [Tajbakhsh and Hassini \(2015\)](#) combine DEA with linear optimization to compare the sustainability performance of different supply chain design choices. [Jakhar \(2015\)](#) incorporates environmental criteria (energy consumption, level of secondary input, carbon footprint) and social criteria (employee development, charity, contribution to community) in a procurement-network-flow problem which is solved combining of fuzzy AHP and optimization. [Varsei et al. \(2014\)](#) stress how the results of an AHP-based supplier selection model can be introduced into a multi-objective optimization covering all sustainability dimensions.

Analytical models, mainly applied by means of adapted versions of the economic order quantity (EOQ) model, derive an optimum for operational problems in the context of Sustainable Warehousing and Eco-efficient Manufacturing. In the field of production planning, [Alinovi et al., 2012](#) present a stochastic EOQ model in a reverse logistics environment deciding upon manufacturing and remanufacturing activities, without explicitly referring to ecological or social indicators. A sustainable EOQ is presented by [Battini et al. \(2014\)](#) by incorporating monetarized

impacts of emissions and waste disposal. For warehousing applications, Bouchery et al. (2012) present a multi-objective EOQ solving a trade-offs between transportation of large batches reducing the transportation-induced carbon emissions but increasing injury risks, and lengthy storing of large quantities that need cooling but expose lower injury rates. The authors implement interactive feedback loops with decision-makers, iteratively revealing their preferences by presenting various objective values differing with regard to weighting coefficients.

6. Discussion & conclusion

This research develops a holistic decision-support framework for corporate SSCM with specific attention paid to the integration of ecological and social indicators into quantitative models. In this way, we address the current scarcity of comprehensive sustainability integrating all three sustainability dimensions. Tackling the research questions, the framework mainly serves two purposes: On the one hand, the structured presentation of concepts and various methods reveals avenues for advancing the research field. On the other hand, the framework proposes a comprehensive toolset for the integration of sustainability into the realm of supply chain management practice. In the following we delineate key benefits for both academia and practitioners.

6.1. Insights for researchers

The evaluated literature provides a number of insights concerning the state-of-the-art of SSCM research. The academic interest on SSCM is on the rise with a growing number of publications explicitly positioned within the research field. The research field significantly evolved since the review compiled by Ashby et al. (2012) with an increased fraction of publications presenting quantitative models. This underlines that the research field is gradually moving to a mature level. Nevertheless, we identified manifold potentials for advancing respective research. First of all, we echo recent publications, such as Brandenburg et al. (2014) and Mota et al. (2015), conceding a dearth of quantitative decision-support comprehensively integrating the social dimension. Indeed, we show that such indicators are only rudimentary treated, if treated at all. Thus, we have to affirm an observation made a decade ago: “The people part is notably absent from OM [Operations Management] research” (Kleindorfer et al., 2005, p. 490). Advancements in Social LCA may yield significant advancements concerning this issue. Secondly, LCAs are frequently named as one of the most common techniques for generation of input parameters in mathematical models and some articles are combining LCA and multi-objective optimization. We assume further potential in this symbiosis. Third, concepts well elaborated in adjacent disciplines did not fully disseminate into SSCM. Further benefits can be realized by integration of concepts like Industrial Symbiosis and by interdisciplinary collaboration with disciplines such as human resource management or environmental management. Apart from these points, this article contributes to SSCM research by clarifying definitions, providing a comprehensive taxonomy of SSCM and drawing a picture of corporate sustainability issues along the value

chain. Although discussable, this article introduces a distinction between the paradigm of SSCM, relevant concepts, and methods and relates these elements to each other within a hierarchical order. To the best of our knowledge such a structured understanding does not exist hitherto and concepts as well as methods for operationalization of SSCM are frequently intermingled in the academic discourse (Taticchi et al., 2015). Concluding, the proposed framework may serve academia as a starting point for advancing SSCM research and maps the field in a broad and integrated manner.

6.2. Insights for practitioners

Steering SSCM-oriented decision-making with the help of the proposed framework, benefits corporate executives in numerous ways. Answering the research questions, this study extends previous works by providing a holistic and integrated depiction of concepts and methods that facilitate operationalizing the sustainability paradigm spanning major supply chain activities. This removes blind spots decision-makers may encounter in the sustainability challenge and eases decision-making in this complex problem area. Practitioners may consult the proposed framework for answering some basic questions within the struggle for sustainability (Table 3). In this way, the framework provides practitioners with a necessary toolkit to prepare decisions in a structured and comprehensive manner. Building on research from different industry backgrounds and using generic reference processes for decision-making, the proposed approach is applicable to various problems decision-makers face when dealing with sustainability issues. This comprises different industry branches as well as problems arising at different points of the value chain. Overall, the framework guides executives towards sustainable business practice and, thus, represents a gradual step heading to a sustainable economy.

6.3. Limitations and conclusion

This framework proposes a generic planning process, which is predominantly based on an extensive literature review. In consequence, two pathways for further validation this research exist. First, the results need to be flanked by action research. Such case based research may challenge some of the assumptions made in this study and may reveal additional insights increasing the applicability of the framework in the real-world. Second, in-depth evaluations of the framework within various industry branches are compulsory and may result in industry-specific adaptations. This task can either be carried out by case studies or focused literature reviews. Moreover, an inherent critique of research based on extant literature may be the question on the completeness of the literature basis. Although following a rigor research process, both the selection and the categorization of the literature as well as structuring of the decision-making framework remains a subjective decision of the authors.

Summing up, we present a comprehensive framework for guiding decision-processes in SSCM contributing to the research field by providing a structured integration of concepts and methods utilizable in the pursuit of ecological and social improvements along the value chain.

Table 3
Contribution of framework to corporate SSCM.

Basic question	Phase of framework	Tool presented in this research
What are the relevant issues along the corporate value chain?	Phase I	SSCM concepts depicted in Fig. 4
What to measure within the struggle for sustainability?	Phase II	Environmental and social indicators summarized in Table 2
How to measure these indicators (Section 5.2)?	Phase II	Data acquisition methods described in 5.2
How to integrate this information into a decision-support model (Section 5.3)?	Phase III	Quantitative methods described in 5.3

Appendix A. Articles providing Quantitative decision-support

Article providing quantitative decision-support	Sustainability Dimension			Focused paradigm	Used modelling technique
	Economic	Ecological	Social		
Alinovi et al., 2012	x			GSCM	Analytical model (EOQ)
Azadi et al. (2015)	x	x	x	SSCM	DEA
Battini et al. (2014)	x	x		None	LCA + Analytical model (EOQ)
Besiou et al. (2012)	x	x	x	CLSC	Single objective Simulation
Borchardt et al. (2011)	x	x		None	Ecological Footprinting + Input-Output-Analysis
Bouchery et al. (2012)	x	x	x	SSCM	Carbon Footprinting + Analytical model (EOQ)
Boukherroub et al. (2015)	x	x	x	SSCM	MOLP
Chaabane et al. (2012)	x	x		SSCM	LCA + MOLP (Optimization)
Choi and Chiu (2012)	x	x		SSCM	Single objective Analytical model (Newsvendor)
Devika et al. (2014)	x	x	x	SSCM	MOLP (Optimization)
Diabat and Al-Salem (2015)	x	x		SSCM	SOLP
Geldermann et al. (2007)	x	x		SSCM	MFA + MCDM (Pinch Analysis)
Govindan et al. (2013)	x	x	x	SSCM	MCDM (TOPSIS)
Govindan et al. (2015)	x	x		SSCM	MOLP (Optimization)
Hsueh (2015)	x	x	x	SSCM	MOLP (Variational Inequality)
Huang et al. (2009)	x			SSCM	Simulation
Jakhar (2015)	x	x	x	SSCM	Carbon Footprinting + AHP + MOLP (Optimization)
Ji et al. (2015)	x	x		GSCM	Single-objective Analytical model (Game theory)
Kostin et al. (2015)	x	x		SSCM	LCA + MOLP (Optimization)
Kuo et al. (2010)	x	x	x	GSCM	AI (ANN) + MCDM (DEA + ANP)
Li (2013)	x	x		SSCM	MOLP (Optimization)
Li and Li (2016)	x	x		SSCM	Analytical model (Game theory)
Mazhar et al., 2007	x			SSCM	AI (ANN)
Mota et al. (2015)	x	x	x	SSCM	LCA + MOLP
Nagurney and Yu (2012)	x	x		SSCM	Analytical model (Game theory)
Neumüller et al. (2015)	x	x	x	SSCM	MCDM (ANP)
Pishvaei et al. (2014)	x	x	x	SSCM	LCA + MOLP (Optimization)
Rager et al. (2015)	x	x		SSCM	MOLP (Optimization)
Rybicka et al. (2015)	x	x		None	MFA
Sitek and Wikarek (2015)	x	x		SSCM	SOLP
Soysal et al. (2015)	x	x		SSCM	SOLP
Su et al. (2016)	x	x	x	SSCM	AI (Grey theory) + MCDM (DEMATEL)
Tajbakhsh and Hassini (2015)	x	x	x	SSCM	MCDM (DEA) + SOLP (Optimization)
Tseng and Hung (2014)	x	x	x	SSCM	SOLP
Validi et al. (2015)	x	x		SSCM	MOLP (Optimization) + MCDM (TOPSIS)
Yung et al. (2012)		x		None	LCA

Notes: SOLP = Single-objective linear program; MOLP = Mixed-objective linear program.

Appendix B. Articles providing qualitative decision-support

Article providing qualitative decision-support	Sustainability Dimension			Focused paradigm
	Economic	Ecological	Social	
Abbasi and Nilsson (2012)	x	x		SSCM
Ageron et al. (2012)		x		SSCM
Amini and Bienstock (2014)	x	x	x	CSR
Azevedo et al. (2012)	x	x	x	SSCM
Beske and Seuring (2014)	x	x	x	SSCM
Fabbe-Costes et al. (2014)		x		SSCM
Fresner and Engelhardt (2004)		x		None
Giannakis and Papadopoulos (2016)	x	x	x	SSCM
Gopalakrishnan et al. (2012)	x	x	x	SSCM
Jabbour and de Sousa Jabbour (2016)	x	x		GSCM
Keating et al. (2008)			x	SSCM
Kjaerheim (2005)		x		Cleaner Production
Koplin et al. (2007)	x	x	x	SSCM
Lake et al. (2015)		x		GSCM
Leigh and Li (2015)	x	x		SSCM
Matos and Hall (2007)	x	x	x	SSCM
Maxwell and van der Vorst (2003)	x	x	x	GSCM
Maxwell et al. (2006)	x	x	x	None
Sahamie et al. (2013)	x	x	x	SSCM
Schrettle et al. (2014)	x	x		GSCM
Seuring (2004)	x	x		SSCM

(continued)

Article providing qualitative decision-support	Sustainability Dimension			Focused paradigm
	Economic	Ecological	Social	
Tsoufas and Pappis (2006)		x		CLSC
Turker and Altuntas (2014)	x	x	x	SSCM
Varsei et al. (2014)	x	x	x	SSCM
Zhu and Cote (2004)	x	x		GSCM

Appendix C. Articles on theory development

Articles on theory development	Sustainability Dimension			Focused paradigm
	Economic	Ecological	Social	
Beske (2012)	x	x	x	SSCM
Boons et al. (2012)	x	x		SSCM
Brindley and Oxborrow (2014)	x	x		SSCM
Carter and Rogers (2008)	x	x	x	SSCM
Ciliberti et al. (2008)		x	x	SSCM
de Brito et al. (2008)	x	x		SSCM
Diabat et al. (2014)	x	x	x	SSCM
Foerstl et al. (2015)	x	x	x	SSCM
Formentini and Taticchi (2016)	x	x	x	SSCM
Gavronski et al. (2011)	x	x		GSCM
Giunpero et al. (2012)	x	x	x	SSCM
Grosvold et al. (2014)	x	x	x	SSCM
de Sousa Jabbour et al., 2014	x	x		GSCM
Khalid et al. (2015)	x	x	x	SSCM
Leppelt et al. (2013)	x	x	x	None
Liu et al. (2012)	x	x		SSCM
Meixell and Luoma (2015)	x	x	x	SSCM
Mitra and Datta (2014)	x	x		SSCM
Moore and Manring (2009)	x	x		SSCM
Morali and Searcy (2013)		x	x	SSCM
Pagell and Wu (2009)		x		SSCM
Pagell et al. (2010)	x	x	x	SSCM
Sancha et al. (2016)			x	SSCM
Seuring (2004)		x		SSCM
Seuring (2011)	x	x	x	SSCM
Seuring and Müller (2008)		x	x	SSCM
Signori et al. (2015)	x	x	x	SSCM
Solér et al. (2010)	x	x		SSCM
Tachizawa and Wong (2014)		x	x	SSCM
Tate et al. (2010)		x	x	SSCM
Vachon and Klassen (2006)		x		GSCM
Vurro et al. (2009)	x	x	x	SSCM
Walker and Jones (2012)	x	x	x	SSCM
Walker et al. (2008)		x		SSCM
Wolf (2011)	x	x	x	SSCM
Wolf (2014)	x	x	x	SSCM
Wolf and Seuring (2010)	x	x		SSCM
Wu and Pagell (2011)	x	x		SSCM
Zailani et al. (2012)	x	x		SSCM

Appendix D. Articles on performance measurement

Articles on performance measurement	Sustainability Dimension			Focused paradigm
	Economic	Ecologic	Social	
Ahi and Searcy (2015)	x	x	x	SSCM
Bai et al. (2012)	x	x		SSCM
Chardine-Baumann and Botta-Genoulaz (2014)	x	x	x	SSCM
Chopra and Wu (2016)	x	x		GSCM
Erol et al. (2011)	x	x	x	SSCM
Golicic and Smith (2013)	x	x		SSCM
Gomes et al. (2014)	x	x	x	None
Gotschol et al. (2014)	x	x		GSCM

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Articles on performance measurement	Sustainability Dimension			Focused paradigm
	Economic	Ecologic	Social	
Kushwaha and Sharma (2016)	x	x		GSCM
Nikolaou et al. (2013)	x	x	x	CSR
Ortas et al. (2014)	x	x	x	SSCM
Roth and Kaberberger (2002)	x	x		None

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